Shell Meta Characters

- Meta characters: Characters that have a special meaning
- Wildcards: * ? []
- Redirection: > >> < |
- Command substitution: $()
- Quoting: ‘ “
- Escape character: \
- Logical operators: && ||
- Background execution: &

Job Control

- Ability to manage multiple processes running in a shell session
  - Start processes in the foreground/background
  - Pause/suspend and resume processes
  - Manage multiple processes concurrently

Example

- atul@lubuntu:~$ **sleep 10**
- atul@lubuntu:~$ **sleep 60 &**
- [1] 1150
- atul@lubuntu:~$ **echo "hello there"**
- hello there
- atul@lubuntu:~$ **jobs**
- [1]+ Running sleep 60 &
- atul@lubuntu:~$ **fg**
- sleep 60

- *What are we doing?*
- Start a process in the foreground
- Then start a process in the background
- Then run another process in the foreground
- Check that the background process is still running
- Finally, bring the background process to the foreground

History

- List of previously executed shell commands
- Commands:
- **history**: Display a list of previously executed commands
- **!!**: Execute the last command again
- **!n**: Execute the command with index *n* from history

The **chown** command

- Change ownership of files/directories
- chown user:group filename – Change owner and group
- chown user filename – Change only owner
- Examples:
  - chown user1 example.txt
  - chown user1:group1 example.txt

Assignment

- Create a file cdac.txt
- Change the default permission of the file cdac.txt:
  - owner/user should have read, write, and execute permissions
  - group should only read and execute permissions
  - others should have no permissions

File Access Control List (ACL)

- **Access Control List (ACL)**: More fine-grained permissions that rwx
- Grant/deny permissions to specific users/groups
- View the ACL of a file: *getfacl file1*
- Set ACL: *setfacl -m u:user1:rwx file1* … Give rwx to user1 for file1
- Set ACL: *setfacl -m g:group1:rwx file1* … Give rwx to group1 for file1
- Remove ACL: *setfacl -x u:user1 file1*

Assignment

- Problem: Login as root, create a file 'filewithacl' and apply acl on it in such a way that only user5 is able to read and write in to it. Note: root will do all its work under 'dbda' folder.
- Solution:
- **su root**
- **cd /path/to/dbda**
- **touch filewithacl**
- **setfacl -x u:: filewithacl** # Remove any existing ACLs for other users
- **setfacl -m u:user5:rw filewithacl** # Grant read and write permissions to user5
- **setfacl -m o::--- filewithacl** # Deny all other users read and write permissions

The mount and unmount Commands

- **mount**: Attach a file system stored on a device such as HDD to a directory (also called mount point), so that the user can access it
- Example: sudo mount /dev/sdb1 /mnt/external
- This command mounts the filesystem from /dev/sdb1 to the directory /mnt/external
- sudo **unmount** /mnt/external … To remove the mounted filesystem

The Global Regular Expression Print (grep) Command

- Search for specific patterns (**regular expressions**) in text
- **grep CDAC hello.txt** Search for the word CDAC in hello.txt file
- **ls | grep ct** List files whose name contains ct anywhere
- **ls | grep ^Pi** List files whose name starts with Pi
- **ls | grep es$** List files whose name ends with es

Common grep Options

Option	Purpose	Example
-i	Ignore case sensitivity	grep -i "linux" file.txt
-v	Invert match (Show lines that to not match)	grep -v "error" file.txt
-c	Count number of matching lines	grep -c "success" file.txt
-n	Show line numbers for matches	grep -n "warning" file.txt
-l	Lowercase L … Show only file names with matches	grep -l "failed" *.log
-w	Match whole words only	grep -w "hello" file.txt
-r	Recursive search in directories	grep -r "TODO" /path/to/dir

Examples

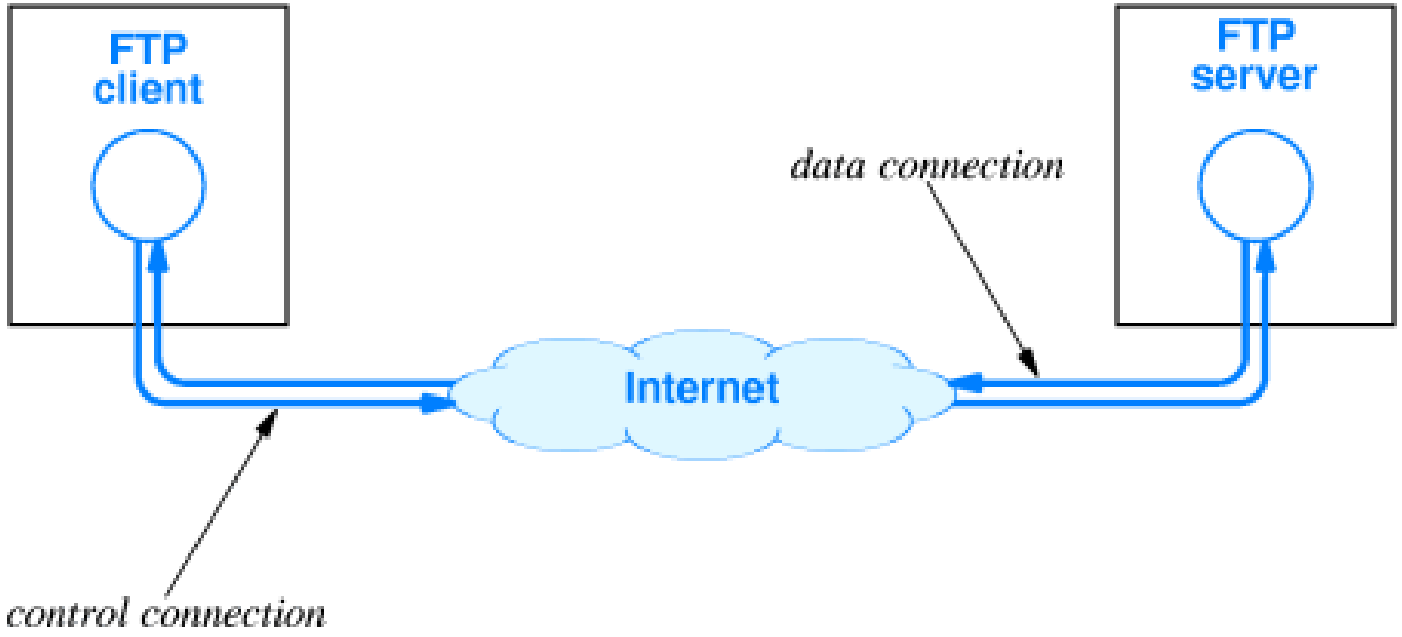
- grep -i "gatsby" greatgatsby.txt
- grep -v "gatsby" greatgatsby.txt
- grep -c "party" greatgatsby.txt
- grep -n "daisy" greatgatsby.txt
- grep -l "Nick" greatgatsby.txt
- grep -w "party" greatgatsby.txt
- grep -r "gatsby" novels/*.txt

find Command

- Search for files and directories
- Find a file based on its name: **find -name one.txt**
- Find all files with extension .txt: **find -type f -name "*.txt“**
- Find all files with size > 10kb: **find -type f -size +10k**
- Same, but also see the actual size > 10kb: **find -type f -size +10k –ls**
- Find directories with permission 755: **find -type d -perm 755 -ls**

FTP, Telnet, etc

FTP Concept


FTP
client
FTP
server
data connection
Internet
control connection

FTP (File Transfer Protocol)

- One of the earliest Internet protocols for file transfers
- Allows downloading (server-to-client) or uploading (client-to-server) of files remotely
- Made up of two parts
  - FTP server: Has files that can be downloaded, or it can also receive files from the users
  - FTP client: Downloads the require files from an FTP server, or it can upload files to the FTP server also

Setting up FTP Server on Ubuntu (on VirtualBox)

- Step 1: A common open-source FTP utility used in Ubuntu is vsftpd.
- **sudo apt install vsftpd**

- Step 2: Start the FTP server
- **sudo systemctl start vsftpd**

- Step 3: Check status
- **sudo systemctl status vsftpd**

- Step 4: Backup configuration files for safety
- **sudo cp /etc/vsftpd.conf /etc/vsftpd.conf_default**

Setting up FTP Server on Ubuntu

- Step 4: Create an FTP user, so that a client can access the FTP server
- **sudo useradd -m testuser**
- **sudo passwd testuser**

- Step 5: Allow FTP through the firewall
- **sudo ufw allow 20/tcp**
- **sudo ufw allow 21/tcp**

- Step 6: Connect to the FTP server locally
- **sudo ftp Ubuntu**
- (Here, Ubuntu is considered to be our Linux machine name, so replace it with the actual machine name … Find it using the command **echo $HOSTNAME**)

Accessing the FTP Server from our Windows Machine

- Open command prompt and type **ftp**
- ftp> **open**
- To 192.168.0.24 … This is the IP address of the Ubuntu machine – to find it, we can type *ip a* or *ifconfig* on the Ubuntu machine's terminal
- Connected to 192.168.0.24.
- 220 (vsFTPd 3.0.5)
- 200 Always in UTF8 mode.
- User (192.168.0.24:(none)): testuser
- 331 Please specify the password.
- Password:
- 230 Login successful.

Accessing the FTP Server from our Windows Machine

- ftp> stat
- Connected to 192.168.0.24.
- Type: ascii; Verbose: On ; Bell: Off ; Prompting: On ; Globbing: On
- Debugging: Off ; Hash mark printing: Off .
- ftp> pwd
- 257 "/home/testuser" is the current directory
- ftp> dir

Checking for Files

- Create a file on the FTP server (i.e. Ubuntu machine)
- atul@atul-virtualbox:~$ **sudo su**
- root@atul-virtualbox:/home/atul# **cd /home/testuser/**
- root@atul-virtualbox:/home/testuser# **nano test.txt**

Now Check in Windows FTP Client

- ftp> pwd
- 257 "/home/testuser" is the current directory
- ftp> dir
- 200 PORT command successful. Consider using PASV.
- 150 Here comes the directory listing.
- -rw-r--r-- 1 0 0 14 Mar 23 19:10 test.txt
- 226 Directory send OK.
- ftp: 69 bytes received in 0.01Seconds 8.63Kbytes/sec.

Download a File using FTP

- ftp> get test.txt
- 200 PORT command successful. Consider using PASV.
- 150 Opening BINARY mode data connection for test.txt (14 bytes).
- 226 Transfer complete.
- ftp: 14 bytes received in 0.00Seconds 14000.00Kbytes/sec.

Telnet and SSH

- **Telnet (Terminal + Network)** is an Internet protocol for allowing a client computer to connect to a remote server computer and work on the server computer
- Somewhat obsolete (outdated)
- Now we generally use **SSH (Secure Shell)** instead of Telnet
- SSH also allows a remote client computer to connect to a server over the Internet
  - Very useful in **cloud computing**

Setting up Telnet Server on Ubuntu

- Step 1: Install Telnet server
- sudo apt install telnetd -y

- Step 2: Check the Telnet status
- sudo systemctl status inetd

- Step 3: Configure firewall for Telnet
- sudo ufw enable
- sudo ufw allow 23

- Step 4: Test
- telnet 192.168.0.24

Accessing Telnet from Windows

- On the command prompt, type telnet
- If you get an error, search for *Turn Windows Features On/Off* and enable *Telnet* there, then retry telnet
- On the telnet prompt, o 192.168.0.24
- User id: testuser, Password: testuser

Setting up SSH on Ubuntu Server

- Step 1: Install SSH server
- **sudo apt install openssh-server**

- Step 2: Check the SSH status
- **sudo systemctl status ssh**

- Step 3: Configure firewall for SSH
- **sudo ufw allow ssh**

- Step 4: Restart if necessary
- **sudo service ssh restart**

Access SSH using Putty

- Download and install **Putty** (https://www.putty.org/)
- Open Putty, enter IP address as IP address of the Ubuntu server (e.g. 192.168.0.24) and connect
- User id: testuser, Password: testuser